

# PHARMACY MANAGEMENT SYSTEM

Project Documentation

**Group 04**



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# 1. Introduction

The Pharmacy Management System is a desktop application built using C# Windows Forms (.NET) with a relational database backend. The system is designed to help a pharmacy digitally manage its day-to-day operations, replacing manual record-keeping with a centralized, role-based application.

The application allows pharmacy staff to manage medicine inventory, categories, suppliers, and customers, process sales through a billing interface, track stock levels and expiry dates, and generate summarized reports. Access to the system is controlled through a login and registration mechanism that supports different user roles, such as Admin and Staff.

This document describes the main functionalities of the application, presents screenshots of each major interface, and outlines the contribution of each group member toward the completion of the project.

## 2. Main Functionalities

The application is organized into a set of forms, each responsible for a specific area of pharmacy operations. The table below summarizes these modules, followed by screenshots of each interface.

| Module               | Description                                                                                                          |
|----------------------|----------------------------------------------------------------------------------------------------------------------|
| Login / Registration | Secure sign-in and new-user (Admin/Staff) account creation with role selection.                                      |
| Dashboard            | Central navigation hub with menu access to Management, Sales, and Reports.                                           |
| Medicine Management  | Add, update, delete, and search medicines with category, supplier, price, quantity, and expiry date.                 |
| Category Management  | Maintain medicine categories such as Tablet, Syrup, and Capsules.                                                    |
| Supplier Management  | Maintain supplier records including contact number and address.                                                      |
| Customer Management  | Maintain customer records including phone number and address.                                                        |
| Billing / Sales      | Create sales invoices by selecting a customer and adding medicines with quantity and price to compute a grand total. |
| Stock Management     | View current stock levels, search medicines, identify low stock, and export stock data.                              |
| Reports Dashboard    | Summarized view of total medicines, customers, suppliers, sales count, and total sales amount.                       |

## 2.1 Login Form

The entry point of the application. Users enter their username and password to log in, with options to navigate to the registration screen or exit the application.



The screenshot displays the login interface of the Pharmacy Management System. The window has a title bar with standard minimize, maximize, and close buttons. The background is a blue-toned image featuring various pharmaceutical icons like a molecular structure, a flask, a syringe, a pharmacy cart, and pills. In the top-left corner, there is a circular logo with a caduceus and the word 'PHARMACY'. A dark blue banner at the top center contains the text 'PHARMACY MANAGEMENT SYSTEM' in white, bold, uppercase letters. Below this, the form includes two input fields: 'User Name' and 'Password', both with yellow backgrounds and blue borders. To the left of each field is a light blue label with the respective text. At the bottom of the form, there are three buttons: a red 'Login' button, a blue 'Sign Up' button, and a red 'Exit' button. A green footer bar at the very bottom contains the text '© 2026 Pharmacy Management System | Developed by Group' in white.

## 2.2 User Registration Form

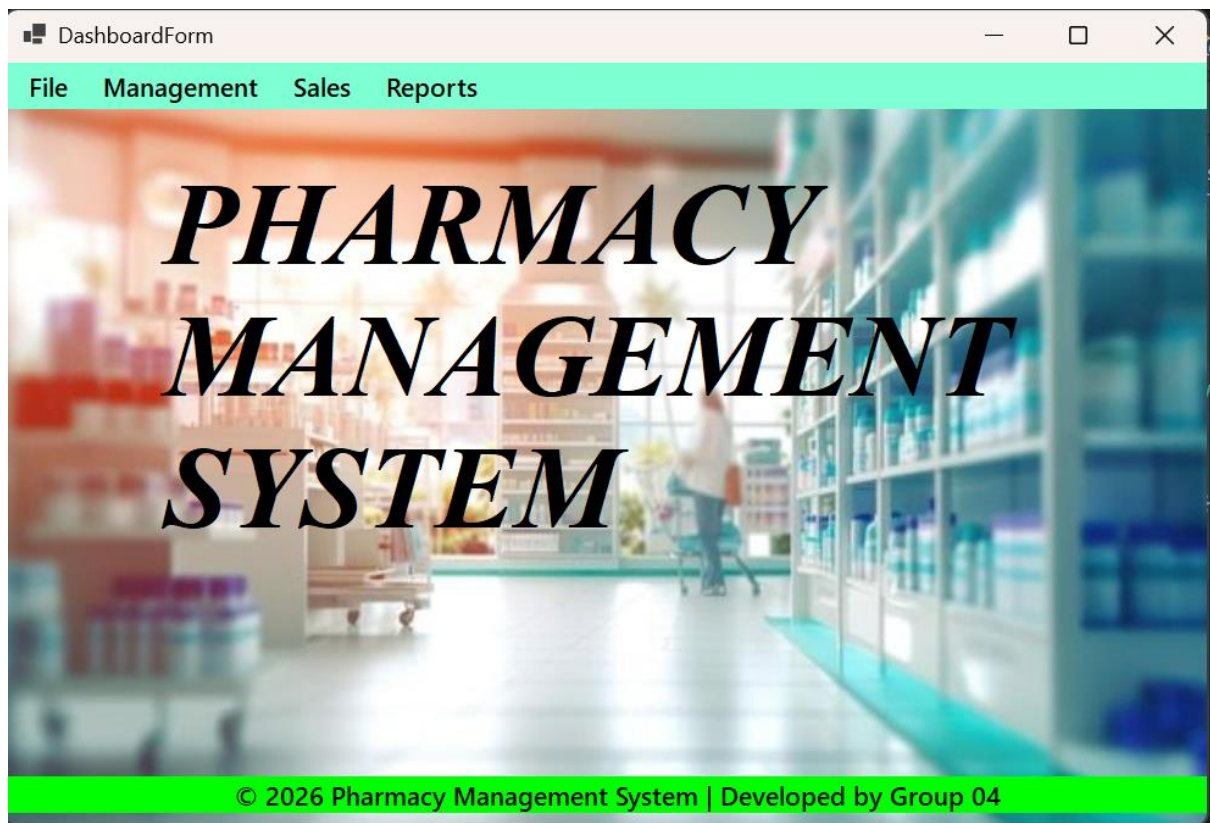
Allows the creation of new system users by capturing their full name, username, password, and assigned role (e.g., Admin), with password confirmation for accuracy.

The image shows a web browser window titled "RegisterForm". The main heading is "PHARMACY MANAGEMENT SYSTEM" in a dark green box. Below it, the section is titled "USER REGISTRATION" in bold blue text. The form contains five input fields: "Full Name:", "Username:", "Password:", "Confirm Password:", and "Role:". The "Role:" field is a dropdown menu currently showing "Admin". At the bottom, there are two buttons: a green "Register" button and a blue "Back" button.

|                                               |                                    |
|-----------------------------------------------|------------------------------------|
| Full Name:                                    | <input type="text"/>               |
| Username:                                     | <input type="text"/>               |
| Password:                                     | <input type="password"/>           |
| Confirm Password:                             | <input type="password"/>           |
| Role:                                         | <input type="text" value="Admin"/> |
| <div><div>Register</div><div>Back</div></div> |                                    |

## 2.3 Dashboard Form

The main hub of the application, providing menu-driven navigation to Management, Sales, and Reports sections after a successful login.



## 2.4 Medicine Management Form

Enables adding, updating, deleting, and searching medicines. Each medicine record includes its name, category, supplier, unit price, quantity, and expiry date, displayed in a sortable grid.



CategoryForm

File

 CATEGORY MANAGEMENT

Category Name :

Add

Update

Delete

Clear

|   | CategoryID | CategoryName |
|---|------------|--------------|
| ▶ | 1          | Tablet       |
|   | 2          | Syrup        |
|   | 3          | Capsules     |
| * |            |              |

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## 2.6 Supplier Management Form

Maintains supplier information, including supplier name, contact number, and address, which is later linked to medicine records

SupplierForm

File


**SUPPLIER MANAGEMENT**

Supplier Name :

Contact No :

Address :

Add
Update
Delete
Clear

Supplier List

|   | SupplierID | SupplierName      | ContactNo  | Email | Address  |
|---|------------|-------------------|------------|-------|----------|
| ▶ | 1          | ABC Pharma        | 0771234567 |       | Colombo  |
|   | 2          | MediPlus Suppl... | 0764567890 |       | Galle    |
|   | 3          | Sun Pharma La...  | 0707890123 |       | Kelaniya |
| * |            |                   |            |       |          |

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## 2.7 Customer Management Form

Maintains customer records, including name, phone number, and address, used during the billing process.

Customer Management

File


**CUSTOMER MANAGEMENT**

Customer Name:

Phone Number:

Address:

Add

Update

Delete

Clear

|   | CustomerID | CustomerName       | Phone      | Address  |
|---|------------|--------------------|------------|----------|
| ▶ | 1          | Sanduni Jayasin... | 0754567890 | Colombo  |
|   | 2          | Ishara Perera      | 0707890123 | Negombo  |
|   | 3          | Chamara Silva      | 0748901234 | Kelaniya |
| ✱ |            |                    |            |          |

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## 2.8 Billing Form

Used to process sales transactions. Staff select a customer and add medicines with their unit price and quantity; the system calculates the line amount and grand total before the sale is saved.

Billing

File


**PHARMACY BILLING SYSTEM**

**Customer Details**

Customer Name:

**Medicine Details**

Medicine:

Unit Price:
Quantity:

Amount:

Add Item

|   | Medicine ID | Medicine Name | UnitPrice | Quantity | Amount |
|---|-------------|---------------|-----------|----------|--------|
| * |             |               |           |          |        |

Grand Total:

Save Sale

Clear

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## 2.9 Stock Management Form

Provides a real-time view of current stock levels for all medicines, with the ability to search by name, filter low-stock items, and export stock data for record-keeping.

File

 STOCK MANAGEMENT

Search Medicine:

Search

Current Stock

|   | MedicineID | MedicineName      | Quantity | ExpiryDate |
|---|------------|-------------------|----------|------------|
| ▶ | 1          | Paracetamol 50... | 100      | 20/06/2027 |
|   | 2          | Cough Syrup 1...  | 25       | 11/05/2027 |
|   | 3          | Omeprazole 20...  | 150      | 20/06/2028 |
| * |            |                   |          |            |

Total Medicines: 3

Refresh

Low

Export

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## 2.10 Reports Dashboard

Displays a consolidated summary of key business metrics, including total medicines, total customers, total suppliers, total sales transactions, and total sales amount.



### 3. Contribution of Each Group Member

The development of the Pharmacy Management System was carried out collaboratively by four group members, each responsible for specific forms and modules of the application, as summarized below.

| Student ID  | Module(s) / Form(s) Developed                               |
|-------------|-------------------------------------------------------------|
| PS/2022/250 | Login Form, Dashboard Form, User Management (Register) Form |
| PS/2022/211 | Medicine Form, Category Form                                |
| PS/2022/084 | Supplier Form, Billing Form                                 |
| PS/2022/218 | Stock Form, Customer Management Form, Reports Form          |